DOCKER COMPOSE - STRUCTURE + FLOW

CUBICLE REFERENCE
Pattern > memorization

REQUEST FLOW

compose.yml | SEE THE PATTERN, NOT REPETITION

```yaml
services:
  frontend:
    image: nginx:alpine              # image OR build
    ports: ["8080:80"]               # HOST:CONTAINER
    volumes: ["./frontend:/usr/share/nginx/html:ro"]
    depends_on: [backend]

  backend:
    build: ./backend                 # folder with Dockerfile
    expose: ["5000"]                 # internal only
    environment:
      DB_HOST: db                    # service name = DNS
      DB_NAME: appdb
      DB_USER: appuser
      DB_PASSWORD: ${DB_PASSWORD}    # .env / secret
    depends_on: {db: {condition: service_healthy}}
    healthcheck:
      test: ["CMD", "curl", "-f", "http://localhost:5000/health"]
    restart: unless-stopped

  db:
    image: postgres:16-alpine
    environment:
      POSTGRES_DB: appdb
      POSTGRES_USER: appuser
      POSTGRES_PASSWORD: ${DB_PASSWORD}
    volumes: ["db_data:/var/lib/postgresql/data"]
    healthcheck:
      test: ["CMD-SHELL", "pg_isready -U appuser -d appdb"]

networks: {default: {name: app_net}}   # shared service DNS
volumes: {db_data: {}}                 # persistent data
```

ESSENTIAL COMMANDS

`docker compose up -d`
start stack

`docker compose up -d --build`
rebuild + start

`docker compose ps`
status / ports

`docker compose logs -f`
follow all logs

`docker compose logs -f backend`
one service logs

`docker compose exec backend sh`
shell into service

`docker compose restart backend`
restart one service

`docker compose config`
validate final YAML

`docker compose pull`
pull newer images

`docker compose down`
stop + remove stack

`docker compose down -v`
DANGER: delete volumes

REMEMBER THIS PATTERN

image / build	existing image OR your Dockerfile
ports	host access: HOST:CONTAINER
expose	internal container port
environment	runtime config; secrets out of Git
depends_on	order; healthcheck adds readiness
volumes	persistent data / host files
networks	service names become DNS names

WHEN IT BREAKS

`ps -> logs -> exec -> curl -> config`

localhost trap: backend reaches DB at "db", not localhost

data trap: down -v deletes named-volume data

MENTAL MODEL: YAML -> network + volumes -> services -> service-name DNS -> publish only what users need